CHAPTER 6: MAINTAINABILITY AND MAINTENANCE

In the previous chapters, we showed that it is important for the operator/user to know the reliability characteristics of the item. We also recognised that it is almost impossible for any item to maintain its function forever, as failure and the degradation of performance is inevitable. Thus, for the user it is equally, or even more important to know:

- When and how often maintenance tasks should be performed
- How they should be performed
- How many people will be needed
- What skills they will need and how much training
- How much the restoration will cost
- How long the system will be down
- What facilities and equipment (special and general) will be required.

All the above information is important as it affects the availability and the life cycle cost of the system. One has to apply a scientific discipline to find answers to these questions.

Maintainability is the scientific discipline that studies complexity, factors and resources related to the maintenance tasks needed to be performed by the user in order to maintain the functionality of a system, and works out methods for their quantification, assessment, prediction and improvement.

The maintainability engineering function involves the formulation of an acceptable combination of design features, which directly affect maintenance and system support requirements, repair policies, and maintenance resources. Some physical design features such as accessibility, visibility, testability, complexity and interchangeability affect the speed and ease with which maintenance can be performed.

MEASURES OF MAINTAINABILITY

1. **Maintenance Elapsed-Time**

   The length of the elapsed time, required for the restoration of functionality, called t*ime to restore,* is largely determined at an early stage of the design phase. The maintenance elapsed time is influenced by the complexity of the maintenance task, accessibility of the items, safety of the restoration, testability, physical location of the item, as well as the decisions related to the requirements for the maintenance support resources (facilities, spares, tools, trained personnel, etc). It is therefore a function of the maintainability and supportability of the system.

Mean Time to Repair

One approach for measuring maintainability is through Mean Time to Repair (MTTR). MTTR is the expected value of the item's repair time. With the knowledge of the reliability and maintainability of the sub-systems one can evaluate the maintainability of the system, that is, mean time to repair of the system, $MTTR_s$ (Birolini, 1994).

Assume that the reliability block diagram of the system has a series structure with n items *with no redundancy.* Let $MTTF_i$ and $MTTR_i$ be the mean time to failure and mean time to repair of sub-system *i* in the system. Consider an arbitrarily large operating time *T*. Assuming that the failure rate of the unit is constant, the expected number of failures of unit *i* in during *T* is given by:

$$\frac{T}{MTTF_i} \tag{5.1}$$

The mean of total repair time to repair unit *i* during *T* is given by:

$$MTTR_i \frac{T}{MTTF_i} \tag{5.2}$$

For the whole system, the mean number of failures is given by:

$$\sum_{i=1}^{n} \frac{T}{MTTF_i} \tag{5.3}$$

For the whole system, the mean of total repair time is given by:

$$\sum_{i=1}^{n} MTTR_i \times \frac{T}{MTTF_i} \tag{5.4}$$

Combining equation (5.3) and (5.4), we get the mean time to repair at the system level, $MTTR_s$, as:

$$MTTR_s = \frac{\sum_{i=1}^{n} \frac{MTTR_i}{MTTF_i}}{\sum_{i=1}^{n} \frac{1}{MTTF_i}} \tag{5.5}$$

Assuming constant failure rate, that is,

$$\lambda_i = \frac{1}{MTTF_i} \text{ and } \lambda_s = \sum_{i=1}^{n} \lambda_i \text{ , equation (5.5) can be written as:}$$

$$MTTR_s = \sum_{i=1}^{n} \frac{\lambda_i}{\lambda_s} MTTR_i \qquad (5.6)$$

Example 5.1

The MTTF and MTTR of four sub-systems in a system are given in Table 5.1. Estimate the system level mean time to repair, $MTTR_s$.

Table 5.1 MTTF and MTTR values for a subsystem

Sub-system	MTTF	MTTR
1	200	24
2	500	36
3	340	12
4	420	8

SOLUTION:

Applying equation (5.5), we get:

$$MTTR_S = \frac{\frac{24}{200} + \frac{36}{500} + \frac{12}{340} + \frac{8}{420}}{\frac{1}{200} + \frac{1}{500} + \frac{1}{340} + \frac{1}{420}} \approx 20 \text{ hours.}$$

2. Maintenance Man Hour (MMH)

Although elapsed time is an extremely important maintenance measure, one must also consider the maintenance man-hours, *MMH* (also known as maintenance labour hours). The MMH is an estimate of the expected "spanner-in-hand" time and takes into account all of the maintenance tasks and actions required for each system, sub-system or component recovery. It should be noted that the MMH can be considerably greater than the elapsed time as it is often possible and sometimes even necessary to employ more than one person on any given activity or task.

MAINTAINABILITY DEMONSTRATION

The objective of the maintainability demonstration is to show that the various maintenance tasks can be accomplished in the times allotted to them. Generally, the most important issue is whether the system can be recovered by sub-system (or line replaceable unit – LRU) exchange within the specified times. It is a common requirement that each LRU can be removed and replaced without interfering with any other LRU. Some of the early jet fighters were virtually built around the engine so that, in order to replace the engine, it was not so much a question of removing the engine from the aircraft as removing he aircraft from the engine.

The demonstration is also expected to generate results that can contribute to the whole development process, identifying any remaining deficiencies such as the design of the system and the test equipment, compilation of maintenance manuals, etc. Any maintainability demonstration would involve the following steps:

1. Identify the operation and environmental condition in which the system is likely to be used.
2. Simulate the system failures and perform corrective maintenance action. One should also record the maintenance man-hours required to complete the repair task successfully.

An estimate for the mean time to failure (or mean time to repair or mean time to support) can be directly obtained from the sample mean. That is,

$$\hat{MTTF} = \sum_{i=1}^{n} \frac{t_i}{n} \qquad (12.6)$$

Estimate for the variance of the failure distribution can be obtained from the sample variance, that is

$$s^2 = \sum_{i=1}^{n} \frac{(t_i - \hat{MTTF})^2}{n-1} \qquad (12.7)$$

Confidence Interval

It is always of the interest to know the range in which the measures such as MTTF, MTTR and MTTS might lie with certain confidence. The resulting interval is called a *confidence interval* and the probability that it contains the estimated parameter is called its *confidence level* or *confidence coefficient*. For example, if a confidence interval has a confidence coefficient equal to 0.95, we call it a 95% confidence interval.

To derive a (1-α) 100% confidence interval for a *large sample* we use the following expression:

$$\hat{MTTF} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) \qquad (12.8)$$

Where $z_{\alpha/2}$ is the z value (standard normal statistic) that locates an area of $\alpha/2$ to its right and can be found from the normal table. σ is the standard deviation of the population from which the population was selected and n is the sample size. The above formula is valid whenever the sample size *n* is greater than or equal to 30. The 90%, 95% and 99% confidence interval for MTTF with sample size $n \geq 30$ are given below:

$$90\% \text{ confidence } \hat{MTTF} \pm 1.645 \times \left(\frac{\sigma}{\sqrt{n}} \right) \qquad (12.9)$$

$$95\% \text{ confidence } \hat{MTTF} \pm 1.96 \times \left(\frac{\sigma}{\sqrt{n}} \right) \qquad (12.10)$$

$$99\% \text{ confidence } \hat{MTTF} \pm 2..58 \times \left(\frac{\sigma}{\sqrt{n}} \right) \qquad (12.11)$$

When the number of data is small (that is when n is less than 30), the confidence interval is based on t distribution. We use the following expression to calculate $(1-\alpha)100\%$ confidence interval.

$$\hat{MTTF} \pm t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) \qquad (12.12)$$

where $t_{\alpha/2}$ is based on (*n*-1) degrees of freedom and can be obtained from *t* distribution table (refer appendix).

Example 5.2

A maintainability demonstration test is carried out on 20 parts and the accomplished repair times are shown in Table 5.2. If the target MTTR is 20 hours, check whether the system has achieved the target maintainability using 95% confidence level.

Table 5.2. Recorded repair times form a sample of 20 parts in hours

8	6	12	20	24
12	9	17	4	40
32	26	30	19	10
10	14	32	26	18

SOLUTION:

Since the observed number of data, n is less than 30, we use t-statistic. The MTTR and standard deviation, s, are given by:

$$MTTR = \frac{1}{20}\sum_{i=1}^{20} t_i = 18.45 \text{ hours}, \quad s = \sqrt{\frac{1}{19}\sum_{i=1}^{n}(t_i - MTTR)^2} = 10.06 \text{ hours}$$

From the t-distribution table (see appendix) we get, $t_{\alpha,\, n-1} = 1.729$ ($\alpha = 0.05$, n-1 = 19).

95% upper limit for MTTR is given by:

$$MTTR + t_\alpha \frac{s}{\sqrt{n}} = 18.45 + 1.729 \times \frac{10.06}{4.472} = 22.33$$

Which is greater than 20 hours, which is the target MTTR. Thus the achieved MTTR is significantly greater than the required MTTR and is therefore not acceptable.

ASSIGNMENT 2:(Submit in the next one week)

Question. 1

Time to failure data for 20 car gearboxes of the model *M2000* is listed in Table 12.1. Find:

1. Estimate of failure function and reliability function.
2. Plot failure function and the reliability function.
3. Estimate of MTTF and 95% confidence interval.

Table 12.1. Failure data of gearboxes in miles

1022	1617	2513	3265	8445
9007	10505	11490	13086	14162
14363	15456	16736	16936	18012
19030	19365	19596	19822	20079

Question. 2

Time taken to complete repair tasks for an item is given in Table 12.3. Find the cumulative time to repair distribution and mean time to repair. Find 95% confidence level for MTTR.

Table 12.3

8	53	71	90
30	56	72	92
31	58	74	94
33	59	75	95
35	61	79	97
40	65	81	99
41	67	82	100
44	68	84	103
49	69	85	108
51	70	89	110

MAINTENANCE

According to BS 4778, maintenance can be defined as:
The combination of all technical and administrative actions, including supervision actions, intended to retain an item in, or restore it to, a state in which it can perform a required function.

In other words, all actions, which keep the system running and ensure that it is maintained to an acceptable standard in which it is able to operate at the required levels efficiently and effectively. The objectives of maintenance are to:

1. Reduce the consequences of failure.
2. Extend the life of the system, by keeping the system in a proper condition for a longer time. In other words, to increase the "up" time of the system.
3. Ensure that the system is fit and safe to use.
4. Ensure that the condition of the system meets all authorised requirements.
5. Maintain the value of the system.
6. Maintain reliability and achieve a high level of safety.
7. Maintain the system's availability and therefore minimise production and quality losses.
8. Reduce overall maintenance costs and therefore minimise the life cycle cost.

MAINTENANCE TASK CLASSIFICATION

All users would like their systems to stay in a state of functioning as long as possible or, at least, as long as they are needed. In order to achieve this, it is necessary to maintain the system's functionality during operation, by performing appropriate maintenance tasks. Thus, maintenance task can be defined as a set of activities that need to be performed, in a specified manner, in order to maintain the functionality of the item/system.

Maintenance tasks can be classified into the following three categories:

1. corrective maintenance task
2. preventive (predictive) maintenance task
3. conditional maintenance task

.1 Corrective Maintenance Task

Corrective maintenance task, CRT, is a set of activities, which is performed with the intention of restoring the functionality of the item or system, after the loss of the functionality or performance (i.e. after failure). Figure 5.3 illustrates typical corrective maintenance task activities. The duration of corrective maintenance task, DMT^c, represents the elapsed time needed for the successful completion of the task. Corrective maintenance task is also referred to as an unscheduled or unplanned maintenance task.

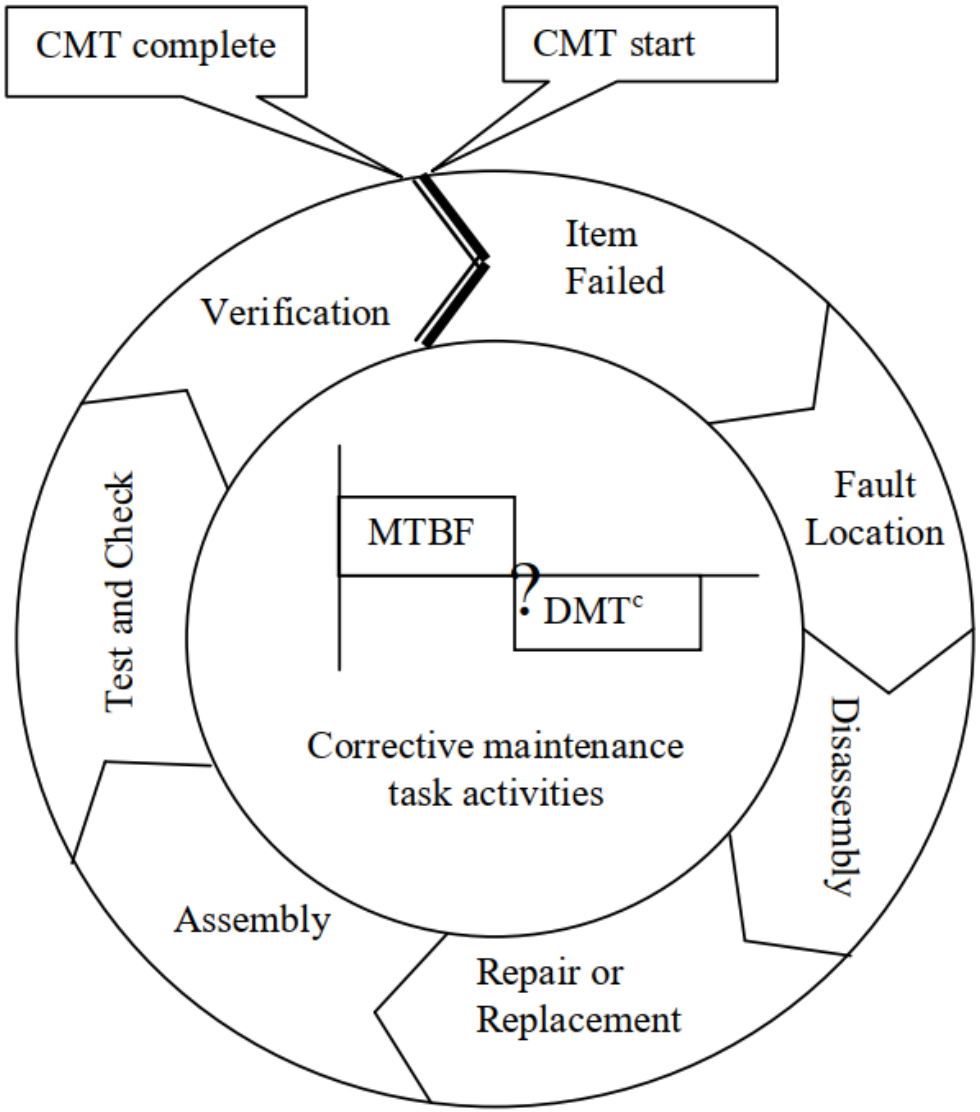

Figure 5.3 Activities of typical corrective maintenance task

.2 Preventive Maintenance Task

Preventive maintenance task, *PMT*, is a maintenance activity that is performed in order to reduce the probability of failure of an item/system or to maximise the operational benefit. Figure 5.4 illustrates the activities of a typical preventive maintenance task. The duration of the preventive maintenance task, DMT^p, represents the elapsed time needed for the successful completion of the task.

Preventive maintenance task is performed before the transition to the state of failure occurs with the main objective of reducing:

- The probability of the occurrence of a failure
- The consequences of failure

Common preventive maintenance tasks are replacements, renewal and overhaul. These tasks are performed, at fixed intervals based on operating time (e.g. hours), distance (e.g. miles) or number of actions (e.g. landings), regardless of the actual condition of the items/systems.

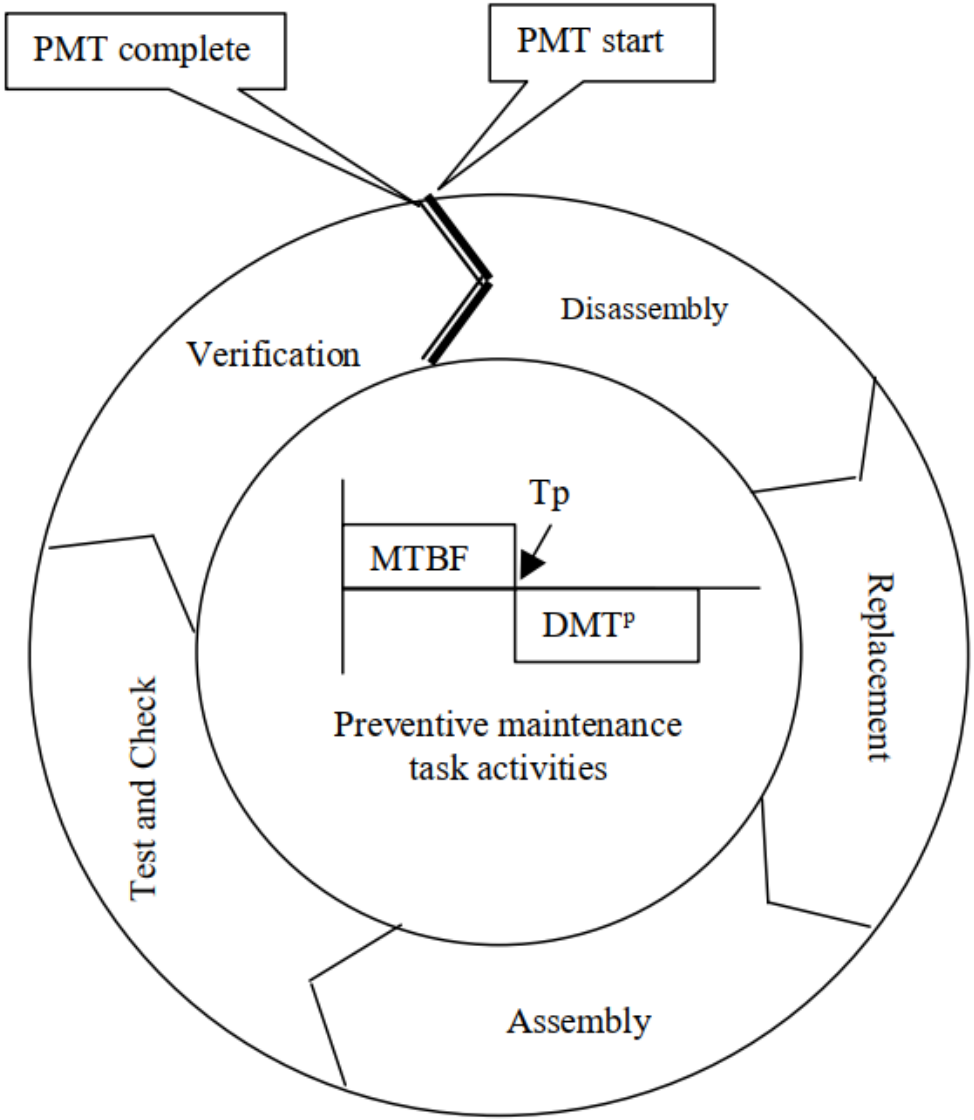

Figure 5.4 Activities of a typical preventive maintenance task

.3 Conditional (Predictive) Maintenance Task

Conditional maintenance task, *COT*, recognises that a change in condition and/or performance is likely to precede a failure so the maintenance task should be based on the actual condition of the item/system. *COT* does not normally involve an intrusion into the system and actual preventive action is taken only when it is believed that an incipient failure has been detected. Thus, through monitoring of some condition parameter(s) it would be possible to identify the most suitable instant of time at which preventive maintenance tasks should take place.

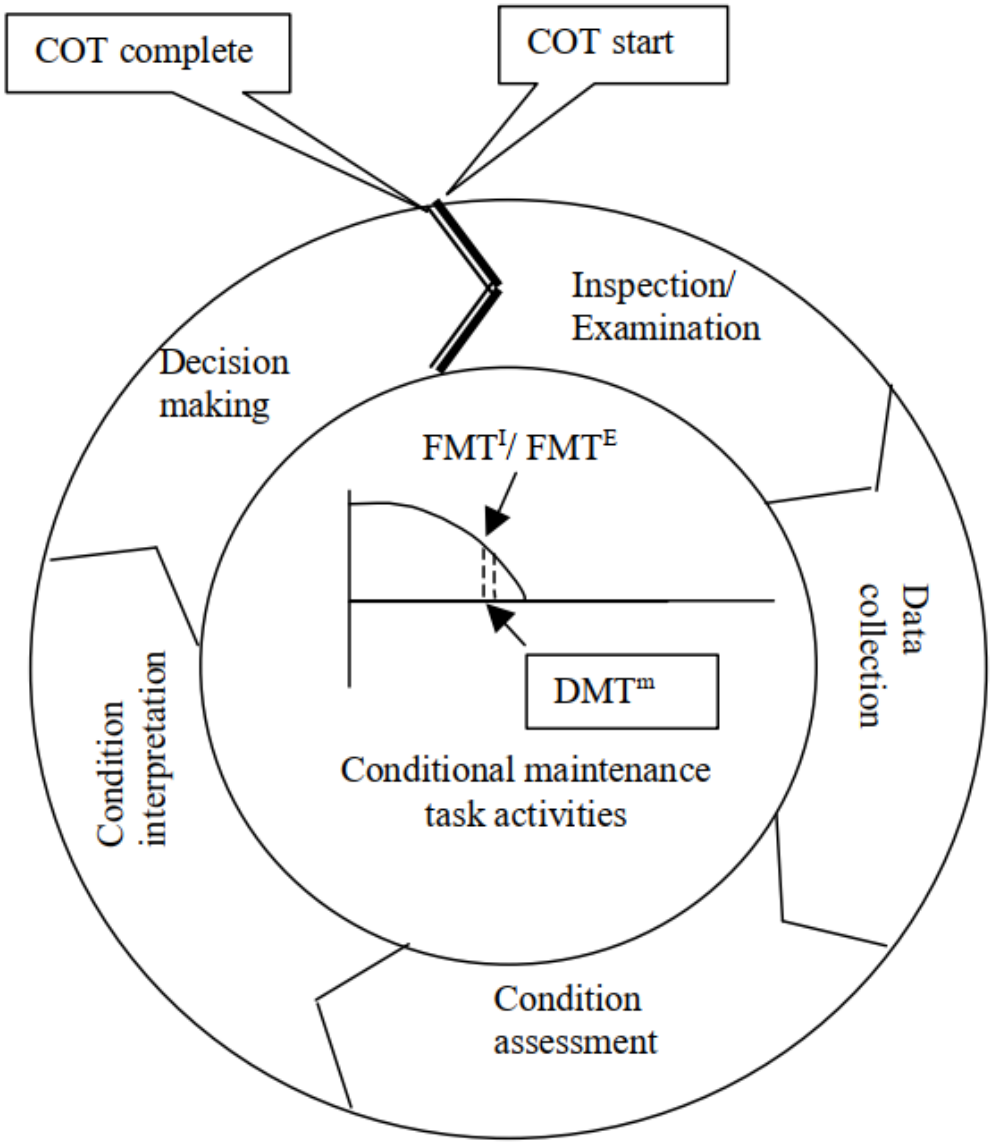

Figure 5.5 Activities of a typical conditional maintenance task.

MAINTENANCE POLICIES

The maintenance policy defines which type of maintenance will (normally) be performed on the various components of the system. It is determined by maintenance engineers, system producers and /or users to achieve high safety, reliability and availability at minimum cost. With respect to the relation of the instant of occurrence of failure and the instant of performing the maintenance task the following maintenance policies exist:

.1 Failure-Based Maintenance Policy

Failure-Based maintenance policy, *FBM*, represents an approach where corrective maintenance tasks are carried out after a failure has occurred, in order to restore the functionality of the item/system considered. Consequently, this approach to maintenance is known as breakdown, post-failure, fire fighting, reactive, or unscheduled maintenance. According to this policy, maintenance tasks often take place in ad hoc manner in response to breakdown of an item following a report from the system user.

Disadvantages of failure based maintenance

Despite the advantages of implementing this policy, it has some disadvantages when it is not correctly selected.

- The failure of an item will generally occur at an inconvenient time.

- Maintenance activities cannot be planned.
- It demands a lot of maintenance resources.
- The failure of an item can cause a large amount of consequential damage to other items in the system.

.2 Time-Based Maintenance Policy

Some failures can lead to economical consequences such as loss of production and therefore a reduction in profit. Some failures may have an impact on the safety of the user, passengers, third parties and environment. Therefore, it is desirable to prevent these failures, if possible, by carrying out maintenance actions before failure occurs.

As the main aim is to reduce the probability of occurrence of failure and avoid the system breakdown, a time-based maintenance policy is performed at fix intervals, which is a function of the time-to-failure distribution of the item considered and in some cases it may be adjusted by the system's user. This policy is very often called age-based, life-based, planned or scheduled maintenance. The reason for that is the fact that the maintenance task is performed at a predetermined frequency, which may be based on, for example, operating times such as, hours, years, miles, number of actions or any other units of use, that make it is possible to plan all tasks and fully support them in advance.

Advantages of time-based maintenance policy

1. Maintenance can be planned ahead and performed when it is convenient from the operational and logistics point of view.
2. The cost of lost production and of consequential damage can be reduced.
3. Downtime, the time that the system is out of service, can be minimised.
4. Safety can be improved.

Disadvantages of time -based maintenance policy

1. Time-based maintenance is performed irrespective of the condition of the system. Consequently, a large number of unnecessary tasks will be carried out on a system that could have been operated safely for a much longer time.
2. The tasks may require higher numbers of skilled mechanics.
3. If the time to perform the maintenance is greater than the time the system would normally be idle (eg overnight) then because of the frequency, it could cause higher levels of unavilability.
4. It cannot guarantee the elimination of all failures and will do nothing to reduce non-age-related failures.
5. Increasing the frequency of maintenance tasks may lead to an increase in the probability of human errors in the form of maintenance-induced failures.
6. Reducing the probability of failure by prematurely replacing components means that the coefficient of utilisation of the item/system, CU^L, will have a value much less than one.

3 Condition Based Maintenance (Predictive Maintenance)

The need for the provision of safety, increased system availability, and reduced maintenance costs have led to an increasing interest in development of alternative maintenance policies. A policy which overcomes many of the disadvantages of the previous maintenance policies (failure-based and time-based), and has proved its ability to extend the operating life of a system without increasing the risk of failure is condition-based maintenance, CBM. CBM is also known as predictive maintenance.

Condition-based maintenance can be defined as: "*Maintenance carried out in response to a significant deterioration in a unit as indicated by a change in the monitored parameters of the unit's condition or performance*" [Kelly & Harris (1978)]. This means that the principle reason for carrying out maintenance activities is the change or deterioration in condition and/or performance, and the time to perform maintenance actions is determined by monitoring the actual state of the system, its performance and/or other condition parameters. This should mean the system is operated in its most efficient state and that maintenance is only performed when it is cost-effective.

The condition-based maintenance is a condition or performance-driven preventive maintenance. This means that the timing of the maintenance task is not simply a function of the mean-time-to-failure. The principle of condition-based maintenance therefore is based on the way of monitoring the condition parameters of a system giving three different types of condition monitoring:

1- Inspection

Inspection is generally performed at regular intervals using any of a number of non-destructive test (NDT) procedures which are designed to determine whether the condition of the (inspected) item is satisfactory or unsatisfactory and hence whether further action is required.

2- Examination

This is a condition-monitoring task, which presents a numerical description of the condition of the item at that moment through relevant condition predictors. The results directly affect the scheduling of the next examination. This is possible because of the unique properties and characteristics of the relevant condition predictor.

3 - Performance Trend Monitoring

For propulsion or energy producing systems, in particular, the "performance" may be expressed as a ratio of the output to input, e.g. miles per gallon, kilometres per litre, thrust per kilogram or (mega)watts per tonne. As the system deteriorates, usually through wear but also through damage, these ratios may show signs of decreasing. For systems operating in relatively constant conditions (e.g. constant ambient temperature, pressure and output), consistent changes in the specific fuel consumption (SFC) will almost certainly be indicative of a deterioration in the system which will need some form of maintenance to restore it to an acceptable level. For systems that are operated in an inconsistent manor for which the environmental conditions may be in a constant state of change, the SFC may be subject to considerable noise and hence any deterioration will only be apparent by using sophisticated trending algorithms, such as Kalman Filtering.

Factor Affecting Maintenance Costs

Maintenance cost could be affected by the following factors:

1. Supply responsiveness or the probability of having a spare part available when needed, supply lead times for given items, levels of inventory, and so on.
2. Test and support equipment effectiveness, which is the reliability and availability of test equipment, test equipment utilisation, system test thoroughness, and so on.
3. Maintenance facility availability and utilisation.
4. Transportation times between maintenance facilities.
5. Maintenance organisational effectiveness and personnel efficiency.
6. Durability and reliability of items in the system
7. Life expectancy of system
8. Expected number of maintenance tasks
9. Duration of maintenance and support task
10. Maintenance task resources

ATTACHMENT 1: T-Distribution Table

ν	α				
	0.100	**0.050**	**0.025**	**0.010**	**0.005**
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.695	6.625
3	1.639	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.626	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.066
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.799
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
∞	1.282	1.645	1.960	2.326	2.576